

The Continuum Hypothesis as a speech act

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March 24, 2026

Abstract

A recent perspective by Feferman (2011) has questioned whether continuum hypothesis (CH) is a well-defined and legitimate mathematical problem. This perspective has received recent criticism by P. Koellner (2016 & 2022) for presenting the risk of reducing mathematics and science to the scope of subjectivism and convention.

In this text, we approach this debate from a linguistic point of view, drawing on pragmatic theories of meaning and their application to mathematics. We focus particularly on speech act theory, and on the hypothesis—defended by Feferman—that mathematical objects function as elements of social reality, in the sense Searle (1995) applies this concept to domains like politics and interpersonal relations. The purpose of this text is to address whether this point is justified, analyzing if speech act theory and Searle’s account of institutions could back up Feferman’s perspective, or at least contribute to a self-consistent account of mathematical reality.

Key words: Continuum Hypothesis, Conceptual Structuralism, Speech Act Theory, Institution Theory.

1 Introduction

In 1883, Cantor, after extending the concept of cardinality to infinite sets, posed the known as Continuum Hypothesis, stating that there could be no strictly intermediate set between the continuum and the set of the integers. After the development of axiomatic set theory, it was shown to be neither refutable (Gödel, 1940) nor provable (Cohen, 1966) by the axioms of ZFC set theory, the ones more widely accepted. Later there have been many ways to try to fix the problem, but neither of them has been successful.

The logical and philosopher of mathematics Solomon Feferman proposed a new and controversial way to look at the problem. He developed a philosophical conception of mathematics called conceptual structuralism, which is based on the idea that mathematical notions originate in processes and activities from real experience that help us understand and model our environment. Around certain considerations, he was led to conclude that CH is an intrinsically vague problem, lacking any mathematical meaning.

This view has not been without its detractors. For example, views such as the ones of Woodin (2001) or Maddy (1997) would directly contradict it, for considering the continuum problem to have a definite truth value, and the main goal of set-theorists to settle it through the introduction of new axioms. But Koellner (2016 & 2022) expressed his disagreement in a much

more direct way. One of the reasons for this disagreement was an important and peculiar feature of Feferman’s conception: its institutional account of the reality and objectivity of mathematical entities. Feferman, in the frame of conceptual structuralism, considers in (2011, p. 10) that mathematical entities’ objectivity lies upon their consistent appearance in independently working individuals; i.e. upon an “intersubjective objectivity that is ubiquitous in social reality”. He considers that mathematical entities exist objectively but are dependent on human constructions, following on this regard Searle (1995, p.1):

[T]here are portions of the real world, objective facts in the world, that are only facts by human agreement. In a sense there are things that exist only because we believe them to exist (...). Things like money, property, governments, and marriages. Yet many facts regarding these things are ‘objective’ facts in the sense that they are not a matter of preferences, evaluations, or moral attitudes.

In any case, he did not explain explicitly the way that mathematical entities could be socially constructed. While in Searle’s account we see quite explicitly, both in (1995) and in (2010), the development of a social institution theory in terms of speech act theory, Feferman simply points out that through a certain social conception of mathematics the truth value of determinate mathematical propositions, including CH, could be indeterminate:

My claim is that the basic conceptions of mathematics and their elaboration are also social constructions and that the objective reality that we ascribe to mathematics is simply the result of intersubjective objectivity (...). That is, it may simply be undecided under a given conception whether a given statement in the language of that conception has a determinate truth value.

Feferman (2011, p. 13)

Maybe the lack of explicitness in this conception of mathematics is what led Koellner to criticize it. In his view, there is a huge difference between how political and social institutions come to exist and how numbers or mathematical entities do. For example, he defends (2022, p. ...) that the reality of the fact that “X is a Spanish citizen” is dependent on institutions such as governments, while the one of “there are infinitely many primes” is not. In general, he considers a huge mistake to make mathematical entities dependent on the intuition or the agreements of a determinate group of people, and that fully undermines, in his view, Feferman’s case on CH.

My main thesis in this text will be that mathematical entities do share, as Feferman points out, many of the features of socially construed entities. In order to defend that, I am going to try to state in precise terms, following in most aspects Searle’s theory of social institutions, how the construction of mathematical entities (and, more precisely, domains) can be understood in a similar way to the establishment of social institutions.³ Once this parallelism has been properly addressed, it will follow in a natural way, I hold, that there are some previous preparatory conditions or presuppositions to the acceptance of a determinate mathematical entity. First, the entities introduced would have to be useful in a systematic way to solve, simplify, or generalize previously defined mathematical or scientific questions. For this reason, often the entities introduced should have a certain correspondence with a previously existent pre-theoretical concept, so that their application in those questions can be more straightforward. These conditions are somewhat coincident, and in any case compatible, with Feferman’s case that CH is an ill-defined mathematical problem, something I will explore in the end of the article. In general terms, I

think that the institutional account for mathematical reality that I am going to explore in this text fits well with many aspects of Feferman's conception of mathematics.

TEXT ORGANIZATION

2 An institutional account for mathematical structures

3 Conclusion

Acknowledgments

Conflict of interest

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